

Thermal Cycling Damage Mechanisms of C/SiC Composites in Displacement Constraint and Oxidizing Atmosphere

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A constraint stress of 62.5 MPa is created on a three-dimensional C/SiC composite specimen whose both ends are fixed when temperature is cycled between 900° and 1200°C. The cyclic stress results in a maximum damage strain of 0.06% within 50 cycles owing to coating and matrix cracking, fiber debonding, sliding, and breaking in the composite. This constrained specimen elongation also leads to a final compressive stress of 14 MPa on the composite through a decrease in the baseline constraint stress. Wet oxygen atmosphere at a high cyclic temperature, concomitant with stresses, can aggravate the damage situation by alternate oxidation between internal and external fibers in composites.

I. Introduction

THE wide range of applications for C/SiC composites includes rocket engine nozzles, thrusters, combustion chambers, divergent/convergent flaps, turbomachinery, and aircraft brakes.^{1–3} In many of the instances under consideration, the composite parts may be subjected to both thermal cycling and rigid constraints in an oxidizing atmosphere. Consequently, thermal cycling behavior of the materials in constrained environments must be understood before actual use.

Thermal stress in a fiber-reinforced light-metal-matrix composite subjected to thermal fatigue and constant strain constraint has been investigated by Chawla.⁴ The static mechanical responses of two- and three-dimensionally reinforced carbon/carbon composites (2D- and 3D-C/Cs) were compared by Aly-Hassan *et al.*⁵ The present paper provides experimental data on the response of a C/SiC composite to thermal cycling in displacement constraint (DC) and wet oxygen atmosphere. Corresponding thermo-mechanical results are analyzed by experimental and theoretical methods. Much of the discussion will then focus on how thermal cycling affects the constrained composite specimens and what damage characteristics can be observed on the tested composites.

II. Experimental Procedure

(1) Preparation of C/SiC Composite

The fiber (T-300™) preform was prepared using a three-dimensional braiding method.[‡] The volume fraction of fibers was about 40%, and the braiding angle was about 20°. Methyl-trichlorosilane (MTS, CH₃ SiCl₃) was used for the infiltration of

the SiC matrix with hydrogen as the carrier gas. Typical conditions for deposition were 1000°C, a hydrogen:MTS ratio of $\alpha = 10$, and a pressure of 4 kPa. Finally, the specimens were cut from the fabricated composite plates and further coated with SiC by I-CVI processing under the same conditions (the residual porosity was about 18% before deposition of the coating and the thickness of the SiC coating was $\approx 50 \mu\text{m}$). The morphology and the dimensions of the as-received specimens are shown in Figs. 1 and 2. The properties of the composites are listed in Table I.

(2) Thermal Cycling Test

Thermal cycling experiments with DC were conducted in an integrated system, which included an induction heating furnace and a servo-hydraulic tester (Model Instron 8801 from Instron Ltd., High Wycombe, UK). As shown in Fig. 3, many experimental conditions/parameters must be taken into consideration, especially regarding (1) the load alignment, (2) the configuration of the specimen, (3) the heater, (4) the cooling water, (5) the infrared pyrometer window, (6) the induction coil for cyclic temperature, (7) the grip holder, and (8) the pressure and flow of the controlled atmosphere. As illustrated by the left y-axis in Fig. 4, thermal cycling was set to run between two selected temperatures ($T_1 \approx 900^\circ\text{C}$ and $T_2 \approx 1200^\circ\text{C}$) with a period of 120 s, namely: heating to T_2 in 60 s and holding for 30 s, and then cooling back to T_1 and holding for 30 s at T_1 ($\Delta T = T_2 - T_1 \approx 300^\circ\text{C}$). Only the middle parts of the specimens (40 mm long, 3 mm wide, and 3 mm thick) were kept in the hot zone and oxidizing atmosphere. The oxidizing atmosphere was either wet oxygen: 7.90 vol% O₂/14.85 vol% H₂O/77.25 vol% argon. The flux of the gas was accurately controlled by a mass flow controller (5850 i series from Brooks, Tokyo, Japan) with a precision of 0.1 mL/min. Strains were measured directly from the

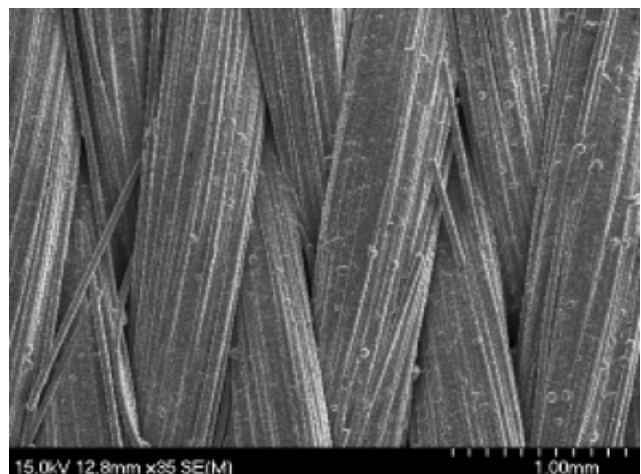


Fig. 1. Scanning electron microscope (SEM) micrograph showing fiber architectures of the fabricated C/SiC composite.

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[‡]3D performs were braided by Nanjing Institute of Glass Fiber, People's Republic of China.

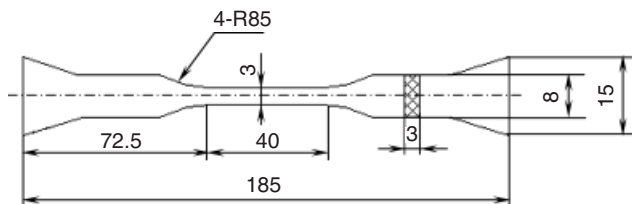


Fig. 2. Drawing of the as-received C/SiC specimen (all dimensions in mm).

gauge length of the specimen by a contact Instron extensometer. The coefficient of thermal expansion (CTE) of the specimens was examined by a dilatometer (Model DIL 402C, Netzsch Ltd., Selb, Germany).

(3) *Description of Constraint Condition*

As shown in Fig. 3, both ends of the specimen were fixed in a pair of wedged steel holders by four bolt fasteners and then the holder was clamped by the crosshead of the tester via a long tensile rod on the other end. The DC used in this investigation was keeping the position of the tester crosshead constant. Thus, longitudinal expansion and shrinkage motions of the specimens were constrained by the holders, and the tester crosshead could detect the constraint-reacting stress from the specimens subjected to thermal cycling. When the specimen is cooled from T_2 to T_1 , constraint stress produced by the holder should be tensile. Conversely, during heating from T_1 to T_2 , the constraint stress will be compressive. The constraint stress can be recorded automatically by an Instron tester. The displacement constraint is abbreviated to DC in the following.

(4) *Measurements and Observations*

Mechanical properties of the specimens before and after cycles were measured with an Instron-8801 tester at room temperature. Coating surfaces and fracture sections of the cycled specimens were observed with a scanning electron microscope (SEM, HITACHI S-4700 and JEOL JSM-6460). In order to estimate damage evolution, thermal cycling tests allow periodic interruptions to obtain the tensile stress-strain curves for calculation of modulus variations of the composites. Thus, the damage indicator D_F can be expressed classically as

$$D_E^N = 1 - \frac{E_N}{E_0} \quad (1)$$

where E_N is the modulus derived directly from the recorded stress-strain loop after N cycles and E_0 is the initial elastic modulus before cycling.

III. Results and Discussion

(1) *Experimental Analysis of Constraint Stress*

Following the test procedure described previously, the constraint stresses of the constrained C/SiC specimens subjected to thermal cycling could be recorded. Figure 4 demonstrates a correlation between the temperature and the constraint stress during thermal cycles by using a magnified view of the abscissa scale. Obviously, the constraint stress changed with thermal cycles number N periodically. Furthermore, the constraint stress

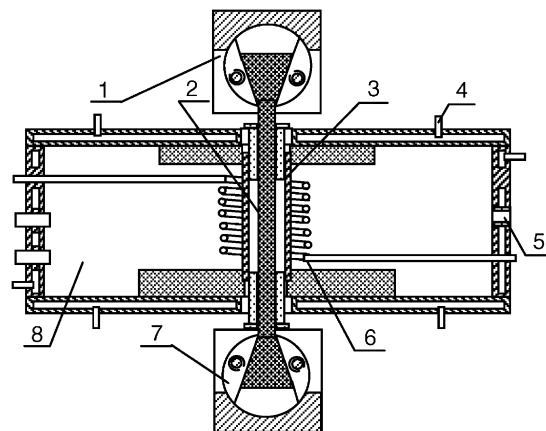


Fig. 3. Schematic drawing of the atmospheric chamber with a detailed view of the grip holders, the furnace, and the specimen (eight major critical points are indicated).

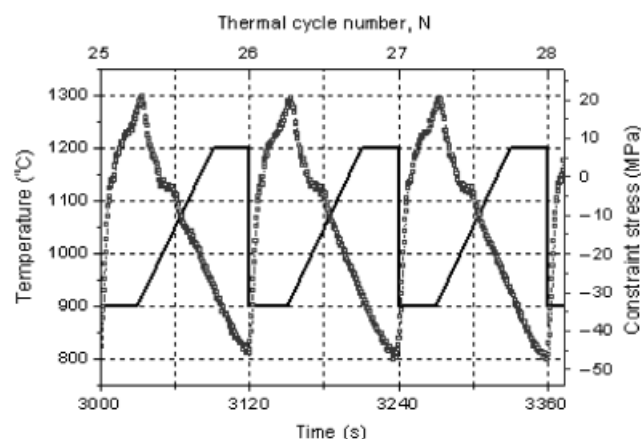


Fig. 4. Correlation between temperature and constraint stress during thermal cycles using a magnified view of the abscissa scale.

increased with cooling from 1200° to 900°C, reaching a peak of 21 MPa, and then decreased upon heating from 900° to 1200°C until a valley of 46 MPa. A distinct inflexion from tension to compression on the constraint stress curve appeared near the zero axis, indicating that the tested specimens were very sensitive to compression stress owing to their thin and long geometric shapes (as shown in Fig. 2). As N increased, saw-toothed stresses repeated periodically with the cyclic temperatures and their periods were just identical to 120 s. An entire constraint stress versus time and N curve of the constrained C/SiC specimen recorded by the Instron tester during thermal cycling (50 cycles, 6000 s) is presented in Fig. 5. Differences (i.e., stress range) between the peaks and the valleys of the stress waves were nearly equal in magnitude and the mean value was about 62.5 MPa. Thermal cycling resulted in cyclic constraint stresses on both ends of the constrained specimens. These cyclic unloading and reloading, similar to external mechanical fatigue loadings, can cause matrix cracking (Fig. 6(a)) transversely, fiber bridging, debonding, sliding, and breaking⁶ (Fig. 6(b)). In addition to the qualitative evidences from Fig. 6, evolution of the damage with increasing N can be assessed quantitatively on the basis of the

Table I. Properties of the As-Received Three-Dimensional C/SiC Composites

Property	Density (g/cm ³)	Modulus (GPa)	Strength (MPa)	Poisson's ratio	Porosity (%)	CTE ($\times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$)			
						900°C	1000°C	1100°C	1200°C
Value	2.0	54	164	0.47	11	3.6415	3.7766	4.3293	4.3719

CTE, coefficient of thermal expansion.

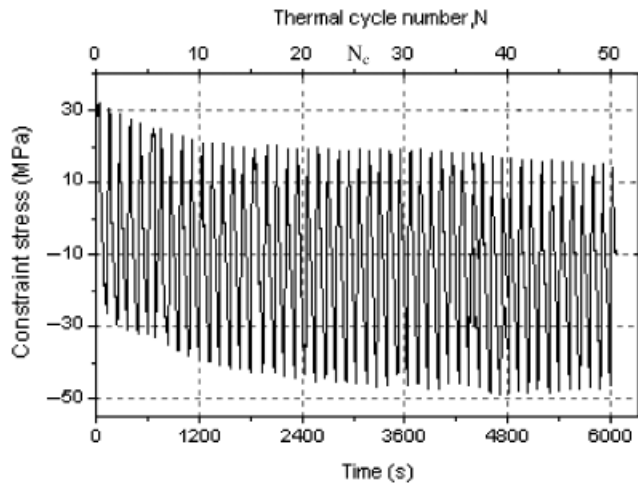


Fig. 5. Typical constraint stress versus time and thermal cycle number curve of the C/SiC specimen recorded by an Instron tester.

damage indicator D_E associated with the modulus variations of the materials. As illustrated by the calculated D_E in Fig. 7, there are two stages in the sequences of damage: (I) a transient increase up to 0.1 within the initial 25 cycles, and (II) a steady state in which the damage tends toward a constant value once N exceeds a critical value N_c of 25. This result is in good accordance with the progression tendencies of the whole constraint stress curves in Fig. 5.

In theory, the strain of the tested specimen under the DC should be constant and the length of the specimen is unchangeable if the long specimen holders are absolute rigid bodies. Actually, in the absence of a strict strain constraint, the long specimen fixed in the long holders is still likely to produce a slight deformation because of the nonreversible damage. It is very interesting in Fig. 5 that an increasing compressive stress occurs on the constraint stress curve with continued cycles on decreasing the central line of the constraint stress curve (i.e., mean stress). The mean stress was found to decrease slightly and gradually down from the initial 0 to the final constant negative value of 14 MPa in average. Simultaneously, the total strain of the specimen, as illustrated in Fig. 8, was shown to increase from the initial 0 to a final nonreversible damage strain of 0.06%, although the value is considerably small. The nonreversible damage strain should include the crack opening displacement, fiber debonding, and sliding mentioned above. It is not surprising that elongation of the constrained specimen resulted in an additional compressive stress. The increase in the compressive stress, however, was limited as thermal cycling damage to the composites could reach saturation when N exceeds the critical value N_c of 25.

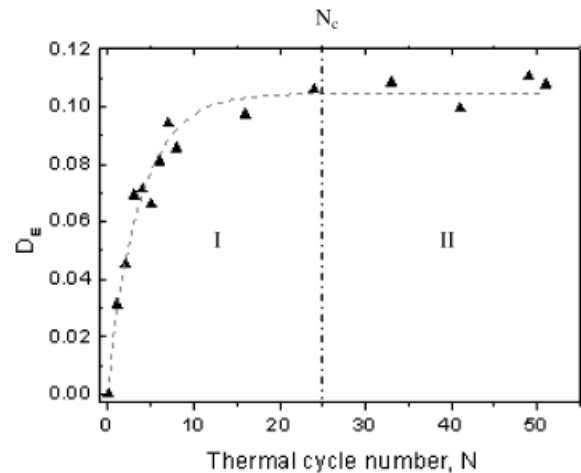


Fig. 7. Damage evolution associated with the modulus variations of the C/SiC composites during thermal cycles.

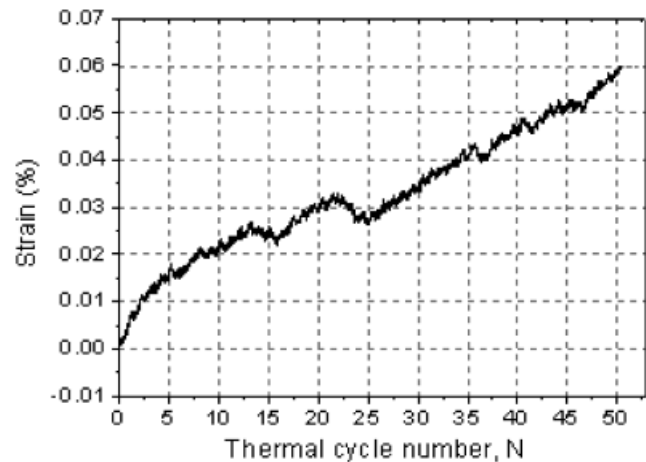


Fig. 8. Strain versus thermal cycle number curve of the C/SiC composite subjected to 50 thermal cycles with DC.

(2) Theoretical Analysis of Constraint Stress

What magnitude is the constraint stress theoretically? Under the cyclic temperature, expansion or shrinkage of the materials is caused by sudden changes in the ambient temperature. Consequently, the constraint stress between two cycled temperatures can be assessed simply as follows:

$$\sigma_{\text{constraint}} = E\alpha\Delta T \tag{2}$$

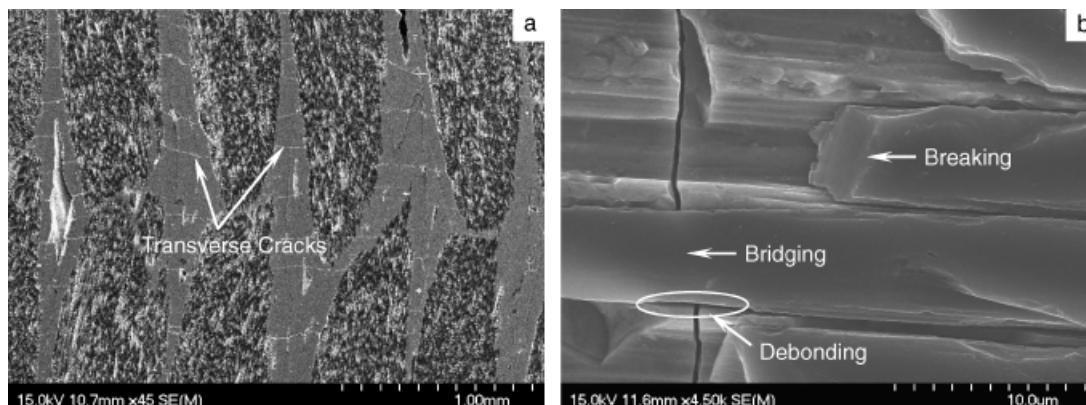


Fig. 6. Scanning electron microscope (SEM) micrograph showing (a) transverse cracks in the matrix and (b) fiber bridging, debonding, and breaking. Transverse cracks marked by arrows in (a) are partially closed once unloading.

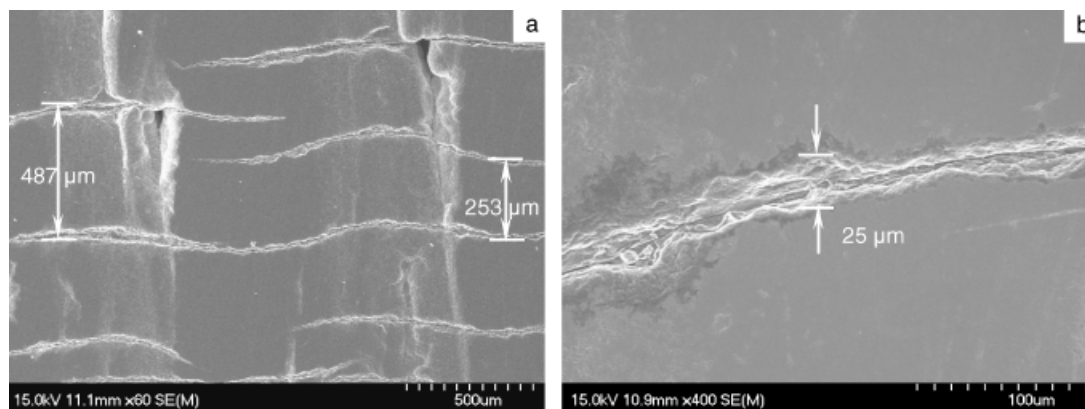


Fig. 9. Micrographs showing (a) wave-shaped coating cracks at a relatively regular spacing and (b) a magnified view of a crushed crack.

where α is the mean linear CTE of the tested specimen and ΔT is the temperature gradient between two selected temperatures.

In the present work, according to Table I, the mean CTE at 900° and 1200°C was $4.0298 \times 10^{-6}/^{\circ}\text{C}$ and the temperature gradient $\Delta T \approx 300^{\circ}\text{C}$. The virgin modulus of the specimens was about 54 GPa. From Eq. (2), the theoretical constraint stress should approximate to 65.283 MPa, which is consistent with the experimental mean value of 62.5 MPa (Fig. 5).

(3) Microstructural Observations

As shown in Fig. 9(a), the outer coatings cracked transversely at a relatively regular spacing (about 253–487 μm). The constraint stress had a significant influence on the crack propagations. On the one hand, the decreasing tensile stress during cycles forced the older and longer cracks to propagate through the whole coating surface while the newer and the shorter ones intercrossed with each other due to lacking driving force. On the other hand, the increasing compressive stress resulted in a crush effect on both sides of a crack, which made the indented cracks thick and rough (about 25 μm as illustrated in Fig. 9(b)). It is noteworthy that the coating regions between two neighboring bundles at the braiding pores were corrugated into wavy shapes. This could be, in large part, attributed to the longitudinal tension effect and the special 3D braided preform structures⁷ (Fig. 1).

An interesting oxidation path around fiber bundles can be observed beneath the coatings in Fig. 10. It can be seen that the oxidation tendencies of fibers both inside and outside around the bundles are nearly identical. When the specimens were cooled from 1200° to 900°C (only 30 s), the outside fibers were oxidized rapidly because of the opening coating cracks created by tensile stress (Fig. 10(a)). Simultaneously, a large

quantity of oxidizing gas easily diffused inward along the propagating cracks and slowly reacted with the inside fibers. Subsequently, the existing cracks were closed and self-sealed upon heating from 900° to 1200 °C because the SiC coating had a larger CTE than the C/SiC substrate and could form a dense silica layer in an oxidizing atmosphere.⁸ Thus, oxidation of the outside fibers stopped temporarily, while the inside fibers, as shown in Fig. 10(b), were likely to be oxidized successively owing to a higher temperature and longer duration (90 s). Nevertheless, fiber oxidation was restrained as the decreasing tensile stress with continued cycles was not high enough to produce extensive coating and matrix cracking leading to fiber exposure. Furthermore, the highly oriented transverse cracks and the increase in compressive stress enhanced the self-healing effect of the existing cracks. As also confirmed by measurements of the residual properties, after 50 cycles, the C/SiC composites underwent a slight loss in the strength and modulus (i.e., about 86.5% and 88.9% of the initial properties, respectively).

IV. Conclusions

Under the DC, constraint stress on C/SiC composite specimens that were cycled between 900° and 1200°C has been measured in good order. The stress was shown to change with cyclic temperature during the same time period of 120 s, namely, it increased during cooling and decreased upon heating. The corresponding stress range between two selected temperatures approximated to 62.5 MPa, which is in agreement with the calculated result of 65.283 MPa. Strain increase in the constrained specimen from 0% to 0.06% resulted in a maximum compressive stress of 14 MPa on decreasing the baseline constraint stress. The damage strain should include crack opening displacement,

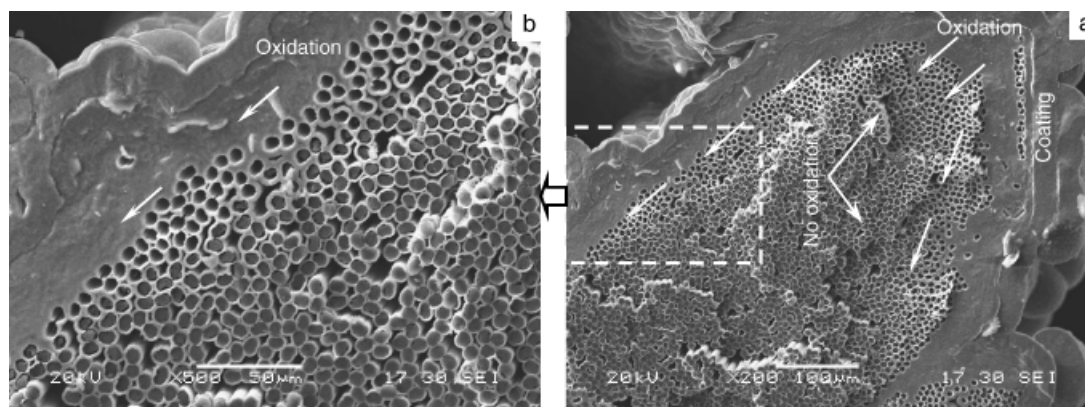


Fig. 10. Typical scanning electron micrograph showing (a) oxidation path of a fiber bundle beneath the coating after 50 thermal cycles and (b) a close-up view of the inside fibers.

fiber debonding, sliding, and breaking. It could be observed that the crushed and wave-shaped cracks intercrossed with each other at a relatively regular spacing on the coating surfaces, and that the internal and external fibers in the same bundles were oxidized alternately along the propagating cracks. As confirmed by the D_E parameter associated with modulus-variations, the microstructural characterization, and the residual properties, under the DC, the C/SiC composites had good damage resistance to thermal cycling and oxidizing atmosphere.

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